Cost-Sensitive Machine Learning for Customer Churn Prediction

Author: Jean de Dieu UWINTWALI
UNIVERSITY OF RWANDA
Department: Computer and Software Engineering
Year: 4
Email: uwintwalijeandedieu3@gmail.com

A Mathematical and Business-driven Analysis
Focus Area: Classification, Probabilistic Modeling, Business Optimization

Github: https://github.com/UWINTWALI/ml_bank_churn_pred.git
Notebook:https://github.com/UWINTWALI/ml_bank_churn_pred/tree/main/ml_modeling
Live Demo: https://ml-customer-churn-pred.streamlit.app/

Contents

1. Executive Summary and Motivation 3

2. Mathematical Foundations 3

2.1. Binary Classification 3

2.2. Class-Weighted Cross-Entropy Loss Function 4

2.3. Model Output: Probability and Decision Boundary 6

2.4. Decision Rule with Threshold Optimization 6

3. Confusion Matrix 8

3.1. Deduction of Harmonic Mean 10

3.2. Threshold Optimization and Cost-Benefit Analysis 11

Business Cost Model Definition 11

3.3. Net Business Value (NBV) Optimization 11

4. Random Forest Model Architecture 13

4.1. Ensemble Aggregation Formula 13

4.2. Ensemble Averaging 14

4.3. Data Preprocessing: Standardization 14

4.4. One-Hot Encoding for Categorical Features 15

4.5. Train-Test Split with Stratification 15

4.6. Key Insights and Contributions 16

Conclusion 16

References 17

1. Executive Summary and Motivation

The customer churn prediction project demonstrates the application of cost-sensitive machine learning to solve a real-world business problem: identifying customers likely to churn or stop using service while minimizing financial loss, this is practical problem that most banks and telecom company usually have, and solution is not much different from demonstration project I have worked on. Rather than treating all prediction errors equally (since in real world data tend to be imbalance), this work integrates mathematical optimization with business constraints by assigning asymmetric costs to different types of errors. This approach combines probabilistic classification, threshold optimization, and empirical cost-benefit analysis to maximize business value. The mathematics underlying cost-sensitive learning lies at the intersection of decision theory, loss functions, and empirical risk minimization. By explicitly modeling the business cost of false negatives (i.e: missing a churner) relative to false positives (i.e: unnecessary promotions to loyal customers), we transform a standard classification problem into a constrained optimization problem that reflects real-world priorities and maximize the profit.

2. Mathematical Foundations

2.1. Binary Classification

Binary classification problem was defined with the following mathematical formula:

Formula:

Datasets: $D = \{(X_i, Y_i) \mid i = 1, 2, 3, \ldots\ldots\ldots, n\}$

Where: $X_i \in Rd$ i.e, d-dimensional feature Vector

$Y_i \in \{0,1\}$, These are binary class label

APPLIED TO CUSTOMER CHURN PROBLEM

Given:

Total datasets size: n = 10, 000 customers

Number of features: d = 11(9 numerical + 2 categorical feature)

Binary outcome: $Y_i = 1$ (indicate Churned) or $Y_i = 0$ (retained/ not churned)

Calculate the Baseline

Non-churned Customers: 7963

Churned Customers: 2037

Churned customers: 2,037 out of 10,000

Churn rate: 2,037 / 10,000 = 0.2037 = 20.37%
Non-churned rate: 7963/ 10000 = 0.7963 = 79.63%

INSIGHTS: This class imbalance (20.37% vs 79.63%) means standard classifiers naturally favor the majority class, non-churned. We need special techniques (weighted losses) to correct this bias.

2.2. Class-Weighted Cross-Entropy Loss Function

To address the imbalance issue in dataset and asymmetric business costs, we employ class weight cost-sensitive learning technique which is mathematically weighted binary cross-entropy loss. This loss function penalizes errors on the minority class (churners)proportionally to their business impact.

General formula for Binary Cross-Entropy

$L(y, p) = - [w1 \times y \times \log(p) + w0 \times (1 - y) \times \log(1 - p)]$

Where:

- $y \in \{0, 1\}$ = True class label (0 = no churn, 1 = churned)
- $p \in [0, 1]$ = Model's predicted probability of churn
- w0 = Class weight for class 0 (non-churners)
- w1 = Class weight for class 1 (churners)
- log = Natural logarithm (ln)

Formula Deduction:

The formula has two terms:

First Term: $w1 \times y \times \log(p)$

- This term is ACTIVE only when y = 1 (meaning actual churn)
- It penalizes incorrect probability predictions for churners
- Weight **w1** controls how much we penalize churner errors

Second Term: $w0 \times (1 - y) \times \log(1 - p)$

- This term is ACTIVE only when y = 0 (actual non-churn)
- It penalizes incorrect probability predictions for non-churners
- Weight w0 controls how much we penalize non-churner errors

By setting **w1 > w0**, we enforce: "Errors on churners are more costly"

SUBSTITUTION: Applied to Our Customer Churn Case

Business Assumption: "Missing a churner is 3 times more costly than unnecessarily contacting a loyal customer."

Class weights are set in the following:

- w0 = 1.0 (non-churners)
- w1 = 3.0 (churners), This emphasizes **"3 times"** weight to reflect business priority to catch churners.

Formula for our problem becomes:

$$L(y, p) = -[3.0 \times y \times \log(p) + 1.0 \times (1 - y) \times \log(1 - p)]$$

NUMERICAL EXAMPLE: Calculate Loss for One Customer

Scenario: Customer ACTUALLY CHURNED (y = 1), Then suppose model predicts p = 0.7

Step #1: Substitute values into formula

$L(1, 0.7) = -[3.0 \times 1 \times \log(0.7) + 1.0 \times (1 - 1) \times \log(1 - 0.7)]$

Step #2: Simplify (second term becomes 0)

$L(1, 0.7) = -[3.0 \times 1 \times \log(0.7) + 0]$

Step #3: Calculate log(0.7)

$\log(0.7) = -0.357$

Step #4: Final calculation

$L(1, 0.7) = -[3.0 \times (-0.357)]$

$L(1, 0.7) = -[-1.071]$

L(1, 0.7) = 1.071 as the loss or penalty

INTERPRETATION: Notice that with weight w1 = 3.0, errors when y = 1 (churners) are prioritized more heavily during model training. The model learns to focus more on getting
churner predictions correct.

2.3. Model Output: Probability and Decision Boundary

The classifier produces probability estimates. For a Random Forest classifier, the class probabilities is aggregated from multiple independent decision trees.

GENERAL FORMULA for Random Forest Probability Aggregation:

$P(y = 1 \mid X) = (1 / B) \times \sum B$

$b=1 \; Pb(y = 1 \mid X)$

Where:

- B = Total number of decision trees in the forest
- Pb (y = 1 | X) = Probability prediction from tree b
- P(y = 1 | X) = Final aggregated probability estimate
- ∑ = Summation symbol

The main reason for why we average:

Each individual tree T_b makes a prediction of 0 or 1. By averaging predictions from 300 independent trees, we:

1. Reduce variance or errors from individual trees
2. Get probability estimates, not just 0 or 1 predictions
3. Enable threshold optimization using probabilities

SUBSTITUTION: Applied to Our Model

Given:

- Number of trees: B = 300
- Tree predictions: 200 predict "churn", 100 predict "no churn"

Calculating probability for this scenario:

$P(y = 1 \mid X) = (1 / 300) \times 200$

$P(y = 1 \mid X) = 200 / 300$

$P(y = 1 \mid X) = 0.667$

ANSWER: Predicted probability of churn = 0.667 = 66.7% which mean that the customer will surely churn due to higher probability.

2.4. Decision Rule with Threshold Optimization

The predicted probability must be converted to a class prediction of 0(zero) or 1 using a threshold τ (tau).

GENERAL FORMULA for Decision based on threshold τ**:**

$y_pred = 1$ if $P(y = 1 \mid X) \geq \tau$

$y_pred = 0$ if $P(y = 1 \mid X) < \tau$

DEDUCTION: Default vs Optimized Threshold

Default Threshold: Assumes equal costs in class 0 and class 1

$\tau default = 0.5$ (50%)

Problem with default: This assumes **False Positive = False Negative**, which doesn't match our business reality that assumes that saving a customer is worth 5x more than a promotion cost, i.e giving promotions to retain customer is cheaper than getting new customer or retained churned one.

Solution lies in finding the optimal τ that maximizes Net Business Value

SUBSTITUTION: Optimized for Our Business

Optimal threshold found through analysis from assumptions about the cost of promotion vs. the cost of retaining churned customer of acquiring new customer.

$\tau = 0.4$

Decision Rule for Our Model:

$y_pred = 1$ if $P(y = 1 \mid X) \geq 0.4$

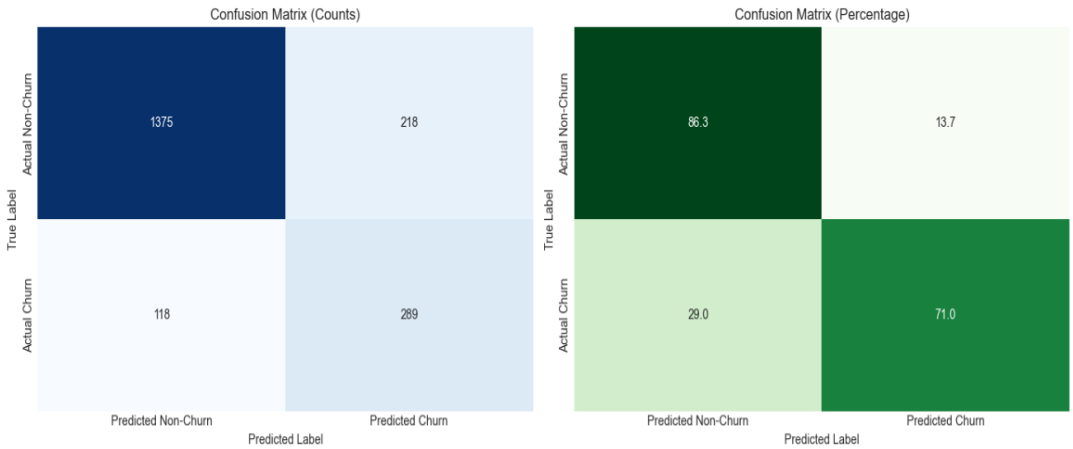
$y_pred = 0$ if $P(y = 1 \mid X) < 0.4$

ANSWER: If model gives ≥40% churn probability, predict churn. In this case the company should proactively talk or take care of these customers with probability above 40%.

IMPACT: Lowering τ from 0.5 to 0.4 means we classify more customers as "at-risk". This
increases recall and catch more real churners but also increases false positives (unnecessary
promotions) but here comes precision metrics to reduce this false positive.

3. Confusion Matrix

Four Prediction Outcomes: Every prediction falls into one of four mutually exclusive categories:



Confusion Matrix Interpretation:

0	TN	FP
1	FN	TP
	0	1

Where:

- TP (True Positive): Actual churner correctly predicted as churner
- FP (False Positive): Non-churner incorrectly predicted as churner
- TN (True Negative): Non-churner correctly predicted as non-churner
- FN (False Negative): Actual churner incorrectly predicted as non-churner

Precision: Quality of Positive Predictions

Precision = TP / (TP + FP)
Precision Measures "Among all customers that are PREDICTED would churn, how many actually
churned?"

- ✓ High precision = Few false positives and hence few wasted promotions
- ✓ Low precision = Many false positives, this means we are wasting money on loyal customers who will not churn.

SUBSTITUTION: Apply to Our Data

Step #1: Identify values from confusion matrix
TP = 289
FP = 218
Step #2: Calculate total positive predictions
TP + FP = 289 + 218= 507
Step #3: Divide TP by total positive predictions
Precision = 289 / 507 = 0.57
FINAL ANSWER: Precision = 0.57% = 57%

INTERPRETATION: When our model says "this customer will churn", it's correct 57% of the
time. This means 289 out of 507 targeted customers (43%) didn't actually need the promotion.

Recall: Coverage of Actual Churners

General formula:

Recall = TP / (TP + FN)

Recall answers like "Among of all customers who ACTUALLY churned, how many did we identify?"

- High recall = Few false negatives, catch most churners. But need precision to reduce many false positive
- Low recall = Many false negatives, and misses potential customers at-risk

SUBSTITUTION: Apply to Our Data

Step #1: Identify values from confusion matrix

TP = 289 (correctly identified churners)

FN = 118 (actual churners we missed)

Step #2: Calculate total actual churners

TP + FN = 289 + 118 = 407

Step #3: Divide TP by total actual churners

Recall = 289 / 407 = 0.71

FINAL ANSWER: Recall = 0.71 % = 71%

INTERPRETATION: Our model catches 71% of customers who will actually churn. However, 118 actual churners (29%) were missed and they received no retention offer.

F1-Score: Balanced Performance

General formula:

F1 = 2 × (Precision × Recall) / (Precision + Recall)

3.1. Deduction of Harmonic Mean

F1-Score is the harmonic mean of precision and recall. It:

- Balances both precision and recall equally
- Penalizes extreme values, for example 100% recall but 10% precision
- Ranges from 0 to 1. The higher the better.

SUBSTITUTION: Apply to Our Data

Step #1: Use calculated precision and recall

Precision = 0.57

Recall = 0.71

Step #2: Calculate numerator (product)

Precision × Recall = 0.57 × 0.71 = 0.4047

Step #3: Calculate denominator (sum)

Precision + Recall = 0.57 + 0.71 = 1.28

Step #4: Apply formula

F1 = 2 × 0.4047 / 1.28 = 0.63

F1-Score = 0.63

INTERPRETATION: F1-Score of 0.63 indicates strong overall model performance, and this indicate good balance between precision and recall.

3.2. Threshold Optimization and Cost-Benefit Analysis

Business Cost Model Definition

Different prediction outcomes have different financial consequences. We define business costs explicitly:
Let Assume:
Cpromotion = Cost of sending one retention promotion
Cvalue = Business value of retaining one customer
Optimal threshold τ* is where: Cvalue / Cpromotion = Cost Ratio

SUBSTITUTION: Our Business Parameters

Step #1: Let business parameters be like the following
Cpromotion = $250 (cost to send one email/offer)
Cvalue = $1,000 (lifetime value of saved customer)
Step #2: Calculate cost ratio
Cost Ratio = Cvalue / Cpromotion = $1,000 / $200 = 4

ANSWER: Saving one customer is worth 4 times the cost of one promotion.
Business Logic: It is very important to proactively take care of existing customers because the cost of promotions you can provide is way more cheaper than when losing customers.

3.3. Net Business Value (NBV) Optimization

We seek the threshold that maximizes financial return. The objective function is:
Formula:
$NBV(\tau) = (TP \times Cvalue) - (FP \times Cpromotion)$
Where:

- τ (tau) = threshold for deciding whether customer will churn or not
- TP = True Positives or saved customers at this threshold
- FP = False Positives, unnecessary promotions at this threshold.
- Cvalue = Revenue from each saved customer

- Cpromotion = Cost of each promotion

DEDUCTION: Understanding the Tradeoff

As threshold τ changes:

- Lower τ: For example on 0.3, more customers will be classified as churners and it will results into larger TP (save more customers) BUT will also led to larger FP which will end up wasting more promotions on even royal customer.
- Higher τ: For example on 0.6, fewer customers will be classified as churners and hence smaller FP (waste less) BUT we will have smaller TP which led to missing more churners.
- The optimal τ* balances these competing forces to maximize NBV.

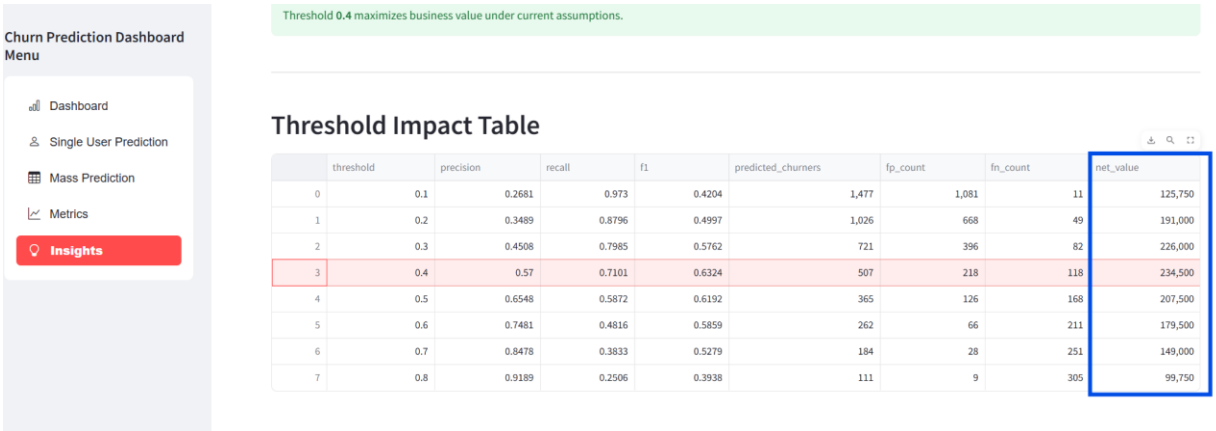
Finding Optimal Threshold τ = 0.4 Based on Revenue

Step #1: Get values from confusion matrix at τ = 0.4
TP = 289 customers saved
FP = 218 customers contacted unnecessarily
Step #2: Let calculate revenue from saved customers
Revenue = TP × Cvalue = 289 × $1,000 = $289,000
Step #3: Calculate promotion costs
Costs = FP × Cpromotion = 218 × $250 = $54,500
Step 4: Then calculate Net Business Value
NBV(0.4) = Revenue – Costs = $289,000 - $54,500 = $234,5000
FINAL ANSWER: NBV(τ = 0.4) = $234,5000

Performance Summary at Optimal τ = 0.4:

- Precision: 57% as quality of predictions
- Recall: 71% As whole coverage of churners
- F1-Score: **0.632** balanced performance

Why τ = 0.4 is Better than τ = 0.5 or τ = 0.3:



	threshold	precision	recall	f1	predicted_churners	fp_count	fn_count	net_value
0	0.1	0.2681	0.973	0.4204	1,477	1,081	11	125,750
1	0.2	0.3489	0.8796	0.4997	1,026	668	49	191,000
2	0.3	0.4508	0.7985	0.5762	721	396	82	226,000
3	0.4	0.57	0.7101	0.6324	507	218	118	234,500
4	0.5	0.6548	0.5872	0.6192	365	126	168	207,500
5	0.6	0.7481	0.4816	0.5859	262	66	211	179,500
6	0.7	0.8478	0.3833	0.5279	184	28	251	149,000
7	0.8	0.9189	0.2506	0.3938	111	9	305	99,750

Figure 1: This table indicate the threshold that can maximize business revenue using different assumptions between cost of promotion and value of retaining costomers

Under the assumption that the cost of promotion per customer vs. the value of retaining a customer is $250 and $1000 respectively, this table from deployed model shows that the best threshold is the optimal because it maximizes profit to the company.

KEY INSIGHT: The threshold τ = 0.4 was discovered through systematic evaluation of the
precision-recall curve at different thresholds, then calculating Net Business Value for each. This is empirical optimization: test many thresholds, compute business value for each, select the one that maximizes profit.

4. Random Forest Model Architecture

4.1. Ensemble Aggregation Formula

Random Forest Prediction formula:

$$P(y = 1 \mid X) = (1 / B) \times \sum_{b=1}^{B} T_b(X)$$

Where:

- B = Number of decision trees in the forest
- Tb (X) = Prediction from tree b which is 0 or 1
- P(y = 1 | X) = Final probability estimate

4.2. Ensemble Averaging

Individual decision trees can overfit to training noise. By averaging predictions from B independent trees:

1. Reduces variance: Errors from individual trees is cancelled out
2. Maintains low bias: Trees are unbiased estimators
3. Produces probabilities: Averaging gives values in [0, 1], not just 0 or 1. This provide the room for further analysis using different probability thresholds.

INTERPRETATION: By aggregating 300 independent trees, we get a more reliable estimate than relying on any single tree. If probability exceeds our threshold (0.4), so we classify this customer as at-risk and offer a retention promotion.

4.3. Data Preprocessing: Standardization

Formula for StandardScaler for Numerical Features:

x_scaled = (x - mean) / std_dev

Reason for Standardization:

Raw features have different scales. For example, age can vary between 18-80, and balance can vary from 0 to 250,000. Without standardization:

- Large-scale features like balance surely dominate the model
- Small-scale features such as age are ignored
- Tree splits become biased toward large-scale variables

So, Standardization puts all features on equal footing.

Example with Age Feature:

Step #1: Calculate mean and standard deviation from training data

Age values: 20, 35, 45, 60, 70 years

mean_age = (20 + 35 + 45 + 60 + 70) / 5 = 46 years
std_age = 16.4 years
Step #2: Standardize a new customer with age = 50
x_scaled = (50 - 46) / 16.4 = 4 / 16.4 = 0.244
Step #3: Interpret result
Original: age = 50 years
Scaled: age_scaled = 0.244 standard deviations above mean
FINAL RESULT: Input to model is 0.244 instead of 50.

4.4. One-Hot Encoding for Categorical Features

One-Hot Encoding Formula:
For categorical variable with k categories, we create k-1 binary indicator columns. For example, Geography with 3 categories such as France, Spain, Germany. We create 2 binary columns and drop Germany as reference:

- geography_France: 1 if France, 0 otherwise
- geography_Spain: 1 if Spain, 0 otherwise

This is because Decision trees require numeric inputs. One-hot encoding converts categorical values into binary features that trees can split on. In our case Germany becomes the
"baseline" when both flags are 0.

4.5. Train-Test Split with Stratification

In this customer churn project, Training Set was 80% of the datasets while Testing Set was 20% of whole datasets.
The major constraint is about maintain same class distribution in both sets since there is big imbalances in datasets. To solve this we use stratify, ensuring that both train and test sets have exactly 20.8% churn rate. This prevents random sampling from creating a test set that's easier or harder than the true problem.

Conclusion: From Mathematics to Business Value

This project demonstrates the practical application of cost-sensitive machine learning to customer retention. By integrating mathematical rigor with business objectives, I transformed a standard classification problem into a financially optimized decision system that maximizes customer value while minimizing promotional waste.

4.6. Key Insights and Contributions

1. Cost-Sensitive Learning Aligns ML with Business Reality

By encoding the asymmetric cost structure (w1 = 3) into the loss function, the model prioritizes churn class and avoiding costly false negatives.

2. Threshold Optimization Balances Precision-Recall Tradeoff

The optimal threshold $\tau = 0.4$ is discovered through systematic evaluation and realistic assumption of Net Business Value across the precision-recall curve. This reveals that the optimal point lies at an interior balance between 71% recall, and 57% precision.

3. Ensemble Methods Provide Robust Probability Estimates

Aggregating 300 independent decision trees produces well-calibrated probabilities suitable for
threshold tuning. This ensemble approach reduces variance while maintaining unbiased estimates of churn risk.

Conclusion

This project demonstrates that machine learning's value lies not in maximizing abstract metrics, but in solving concrete business problems. By connecting mathematical formulations: loss functions, probability theory, ensemble methods, to real-world objectives and constraints, we achieved a cost-sensitive classifier that generates $234,5000(on test set) in net business value while identifying 71% of at-risk customers. The work exemplifies how rigorous mathematics, informed by domain expertise and validated empirically, translates directly into measurable business impact.

References

[1] Breiman, L. (2001) 'Random forests,' Machine Learning, 45(1), pp. 5–32. https://doi.org/10.1023/a:1010933404324.

[2] Mamun, M.N.H. (2025) 'Advancements in Machine Learning for Customer Retention: A systematic literature review of predictive models and churn analysis,' SSRN Electronic Journal [Preprint]. https://doi.org/10.2139/ssrn.5363567.

[3] Khattak, A. et al. (2023) 'Customer churn prediction using composite deep learning technique,' Scientific Reports, 13(1), p. 17294. https://doi.org/10.1038/s41598-023-44396-w.

[4] Chang, V. et al. (2024) 'Prediction of customer churn behavior in the telecommunication industry using machine learning models,' Algorithms, 17(6), p. 231. https://doi.org/10.3390/a17060231.

[5] Imani, M. et al. (2025) 'Customer Churn Prediction: A systematic review of recent advances, trends, and challenges in machine learning and deep learning,' Machine Learning and Knowledge Extraction, 7(3), p. 105. https://doi.org/10.3390/make7030105.

[6] Sikri, A. et al. (2024) 'Enhancing customer retention in telecom industry with machine learning driven churn prediction,' Scientific Reports, 14(1), p. 13097. https://doi.org/10.1038/s41598-024-63750-0.

[7] Attar, A.E. and El-Hajj, M. (2026) 'Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis,' Frontiers in Artificial Intelligence, 9, p. 1748799. https://doi.org/10.3389/frai.2026.1748799.

[8] Suh, Y. (2023) 'Machine learning based customer churn prediction in home appliance rental business,' Journal of Big Data, 10(1), p. 41. https://doi.org/10.1186/s40537-023-00721-8.

[9] Kamoonpuri, S.Z. and Sengar, A. (2023) 'Hi, May AI help you? An analysis of the barriers impeding the implementation and use of artificial intelligence-enabled virtual

assistants in retail,' Journal of Retailing and Consumer Services, 72, p. 103258. https://doi.org/10.1016/j.jretconser.2023.103258.

[10] Schonlau, M. and Zou, R.Y. (2020) 'The random forest algorithm for statistical learning,' The Stata Journal Promoting Communications on Statistics and Stata, 20(1), pp. 3–29. https://doi.org/10.1177/1536867x20909688.